

QUEUE QNO:3

QUESTION:

Question description

Ramesh is an DS expert training youngsters struggling in DS to make them better.

Ramesh usually gives interesting problems to the youngsters to make them love the DS.

One such day Sathya provided to the youngsters to solve the task such that, delete an element in a Queue, Queue data structures work on the FIFO architecture so the element that has entered first in the list will go out from the list first.

Youngsters were lacking the idea to solve the problem.

Being an exciting youngster can you solve it?

every lines indicates the enqueue of each elements

last line indicates the fine queued elements.

Logical Test Cases

Test Case 1

INPUT (STDIN)

```
5
5 6 3 1 2
```

EXPECTED OUTPUT

```
Dequeuing elements:
6 3 1 2
3 1 2
1 2
2
```

Test Case 2

INPUT (STDIN)

```
8
1 8 6 4 9 5 3 5
```

EXPECTED OUTPUT

```
Dequeuing elements:
8 6 4 9 5 3 5
6 4 9 5 3 5
4 9 5 3 5
9 5 3 5
5 3 5
3 5
5
```

Mandatory Test Cases

Test Case 1

KEYWORD

```
void enqueue(int data)
```

Test Case 2

KEYWORD

```
for(i=front;i<=rear;i++)
```

Test Case 3

KEYWORD

```
enqueue(data);
```

Test Case 4

KEYWORD

```
void dequeue()
```

CYCLOMATIC COMPLEXITY

3

TOKEN COUNT

250

NLOC

59

Source Code:

```
#include <stdio.h>
int queue[105];
int front = 0;
int rear = -1;
void enqueue(int data) {
    rear++;
    queue[rear] = data;
}

void dequeue() {
    if (front <= rear) {
        front++;
    }
}

void display() {
    int i;
    for(i=front; i<=rear; i++) {
        printf("%d", queue[i]);
        if (i < rear) {
            printf(" ");
        }
    }
    printf("\n");
}

int main() {
    int size;
    if (scanf("%d", &size) != 1) return 0;
    for (int j = 0; j < size; j++) {
        int data;
        scanf("%d", &data);
        enqueue(data);
    }
    printf("Dequeuing elements:\n");
    for (int j = 0; j < size - 1; j++) {
        dequeue();
        display();
    }
    return 0;
}
```

Output:

```
5
5 6 3 1 2
Dequeuing elements:
6 3 1 2
3 1 2
1 2
2
```

```
8
1 8 6 4 9 5 3 5
Dequeuing elements:
8 6 4 9 5 3 5
6 4 9 5 3 5
4 9 5 3 5
9 5 3 5
5 3 5
3 5
5
```